

ABDULLAH OTHMAN

Ph.D., Electrical Engineering — Radio Resource Management & Wireless Digital Twins

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PROFILE

Ph.D. in Electrical Engineering with research in **radio resource management** and **risk-aware optimization** for next-generation wireless networks — including **digital-twin frameworks** and non-terrestrial (UAV / high-altitude platform) systems. Work covers outage detection and coverage planning under uncertainty, interference management with Kalman-filter-based user tracking, and coverage enhancement for NB-IoT / industrial-IoT networks. Co-inventor on **4 granted U.S. patents** in deep-learning-based event detection. Published across IEEE venues.

EXPERIENCE

Doctoral Researcher

École de technologie supérieure (ÉTS)

 2021 – 2026  Montréal, QC

- Built **digital-twin** frameworks coupling ray-traced channel models with resource-allocation logic to detect outages and plan UAV-assisted uplink coverage for disaster-response networks. *IEEE Comm. Letters*, 2026.
- Designed **radio-resource-management** algorithms for NB-IoT / industrial-IoT coverage enhancement — joint power control, sub-carrier selection, and cell-radius planning, solved at scale for massive-device scenarios. *IEEE TNSM*.
- Built an **OFDMA interference-management** scheme for UAV / high-altitude-platform networks that tracks users with Kalman filtering and keeps co-channel interference safely below link thresholds under position uncertainty.
- Designed an **energy-efficient UAV trajectory-and-power** controller that meets SINR targets under heavy impulsive interference by combining robust user tracking with data-driven interference margins.
- Analyzed convergence of distributed power control in multi-microgrid wireless systems. *IWCMC* 2025.

Electrical Engineer

Naizak Global Engineering Systems

 2018 – 2021  Khobar, Saudi Arabia

- Diagnosed and repaired lab instrumentation (distillers, thermostats, flow/pressure rigs, custom electronic boards) at **PCB and IC level** for KFUPM departments, keeping replacement costs down.
- Prepared technical proposals and coordinated multi-department equipment relocations for the Lab Technical Support Office.

Research & Teaching Assistant

KFUPM

 2015 – 2018  Dhahran, Saudi Arabia

- M.Sc. research on wireless-link design for seismic data acquisition: maximized uplink sum-rate on a multi-access channel using integer programming and metaheuristics (particle swarm, ant colony, simulated annealing). *SEG* 2018.

EDUCATION

Ph.D. Electrical Engineering

ÉTS

 2021 – 2026  Montréal

Thesis: Digital-Twin-Enabled and Risk-Aware Radio Resource Management for Resilient IIoT Smart Grid Communications. Advisor: Prof. Georges Kaddoum. **Mention: Excellent** (highest grade).

M.Sc. Telecommunications

KFUPM

 2015 – 2018  Dhahran

B.Sc. Electrical Engineering

KFUPM

 2011 – 2015  Dhahran

FOCUS AREAS

Radio resource mgmt. 5G/6G NB-IoT
OFDMA Non-terrestrial (UAV/HAPS)
Digital twins Signal processing Optimization

AWARDS



Mention Excellent — Ph.D. thesis
Highest grade conferred by ÉTS, May 2026.



Jury nomination — thesis excellence award
ÉTS Ph.D. defense, April 15, 2026.



University Scientific Forum
1st Place (Dept.) & 4th Overall, KFUPM, 2018



Letter of Commendation — Teaching Excellence
KFUPM Teaching Lab, 2017

FUNDING

Supported by Hydro-Québec & NSERC Alliance Grant (ALLRP 552856-20), 2021 – 2025. Previously supported by CeGP Energy and Geo-Processing Grant (GTEC1601, KFUPM & Georgia Tech), 2016 – 2018.

SKILLS

- ## PUBLICATIONS

Selected Journal Papers

- ### Patents (all granted, U.S.)

- “Computer System for Event Detection of Passive Seismic Data”. US 12,196,902 B2, Issued Jan. 14, 2025.
- “Apparatus and System for Identifying Seismic Features”. US 12,117,577 B2, Issued Oct. 15, 2024.
- “Apparatus with Circuitry for Seismic Data Detection”. US 12,105,236 B1, Issued Oct. 1, 2024.
- “Event Detection and De-noising Method for Passive Seismic Data”. US 12,092,780 B2, Issued Sep. 17, 2024.

Wireless & Signal Processing

5G/6G NR NB-IoT OFDMA Beamforming
Channel modeling Ray tracing Kalman filtering

Optimization

Machine Learning

Deep learning CNNs TensorFlow

Keras

Programming

MATLAB Python C++ $\text{\LaTeX}$

Tools

CVX Gurobi Sionna Mathematica

VOLUNTEERING

2018 Station platform agent, MMSL
Makkah Train (Hajj Season).

2013 EE Consultants Committee, KFUPM – contributed to improving lab courses and introducing new projects.

LANGUAGES

English
Arabic
French

